LESSON 9 • TERM 1 • LEVEL 1

Draw and Measure

Length: centimetres

TODAY WE WILL

Use a ruler to measure and to draw lines exactly.

Hi, I'm Mi. Let's notice, test and explain why!



Warm-up: measure with clips

4–10 MIN

1. Lay clips along your pencil, end to end.
2. No gaps. No overlaps.
3. Count the clips.

WHAT IF?

Your friend uses bigger clips. Will the number be the same?

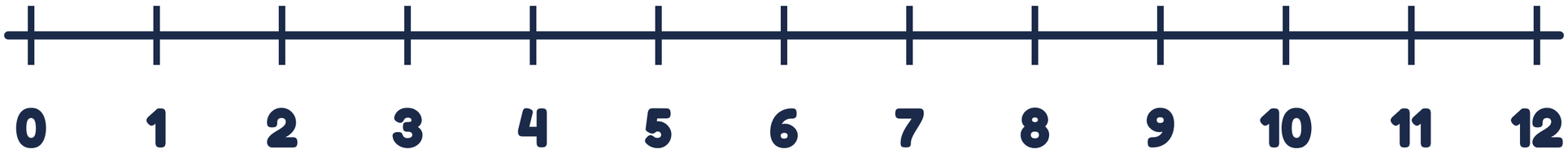


Rulers ready? Off we go!

4–10 MIN

Find 0 on your ruler

Put your finger on the 0 mark. Count along the marks.

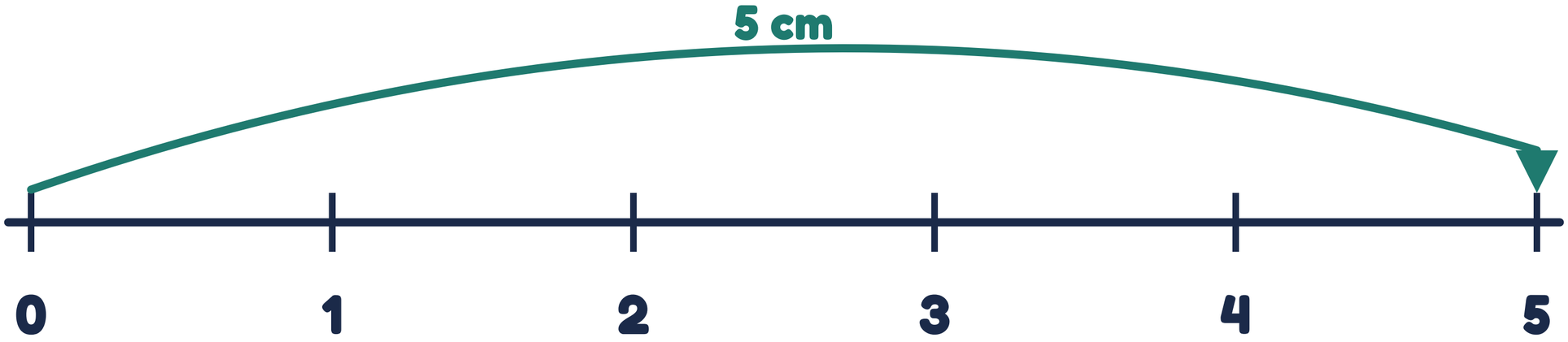


Say the numbers with me

4–10 MIN

One centimetre

The gap between two marks is 1 cm.



Slide your finger from 0 to 5. How many gaps? How many cm?

Sneak-peek puzzles

10–14 MIN

Ideas we have not learnt yet. A guess is a great start. Nobody is marked.

PUZZLE 1: anyone can try

Draw a dot.

Use your ruler.

Draw a line from the dot, exactly 3 cm long.

PUZZLE 2: think

Ali puts one end of a pencil at the 1 mark of his ruler.

The other end is at the 9 mark.

Ali says it is 9 cm long. Is he right? Write yes or no. Then write why.

Explore: guess, then measure

14–28 MIN

1. Find 3 things: a pencil, a spoon, a rubber or a toy.
2. Guess how many cm long each one is.
3. Put one end at the 0 mark. Read the mark at the other end.
4. Say: my ___ is about ___ cm.

WHAT IF?

The end is between two marks. Say: about ___ cm.

Draw it!

14–28 MIN

1. Put a dot where the line starts.
2. Put the 0 mark on the dot.
3. Make a small mark at the end.
4. Join them with a straight line.

3 cm | **8 cm** | **12 cm**

What did we notice?

14–28 MIN

- Where do we start measuring?
- Why do we all use centimetres?
- Why is a ruler better than clips?

The parts of a ruler

28–38 MIN



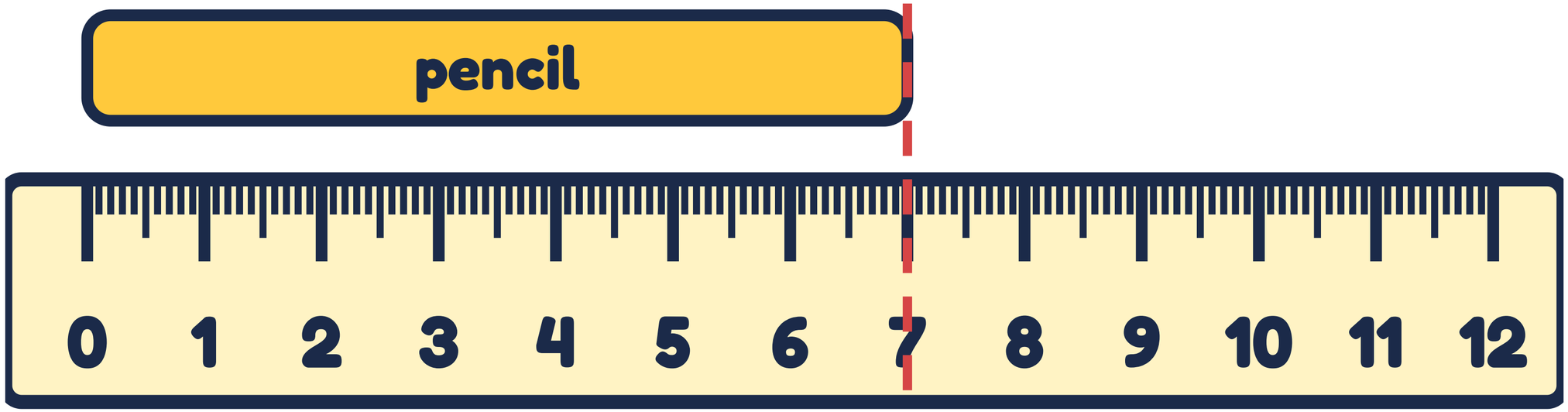
REMEMBER

0 is where we start.

The gap from one mark to the next is 1 cm.

Measure: three steps

28–38 MIN



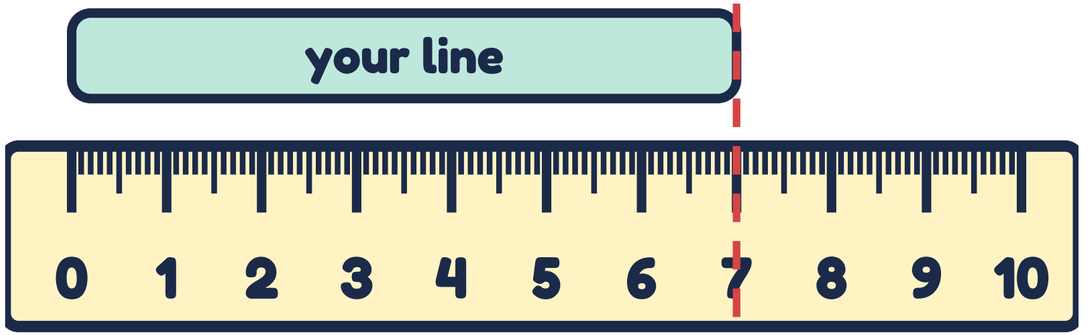
1. Put one end of the thing at the 0 mark.
2. Keep the ruler straight along the thing.
3. Read the mark at the other end. That is the length in cm.

Look from straight above

Draw a line 7 cm long

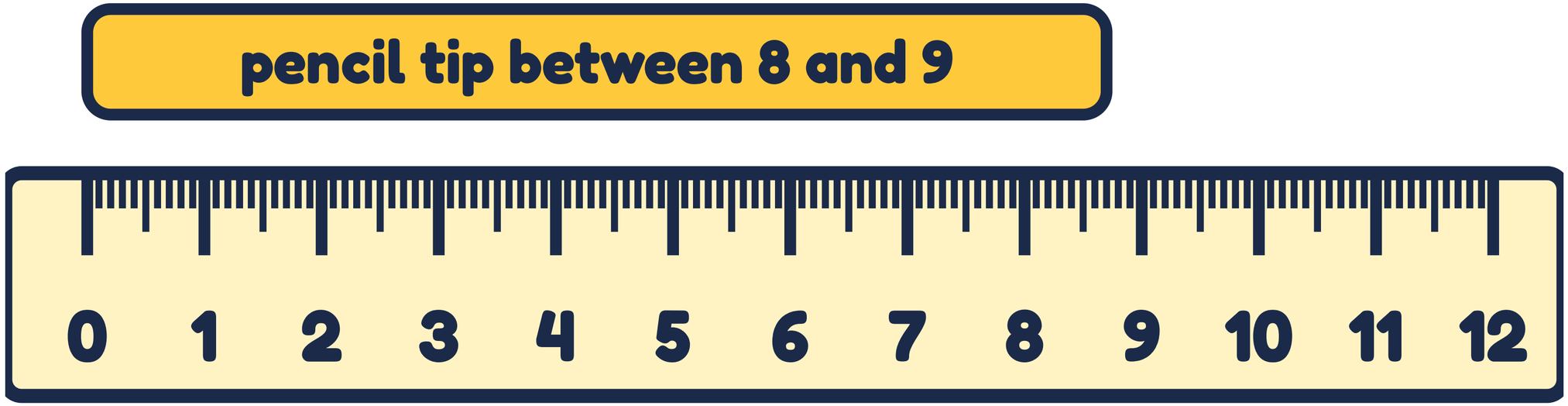
28–38 MIN

1. Dot where the line starts.
2. The 0 mark on the dot.
3. Small mark at the 7.
4. Join with a straight line.
5. Check: 0 at the start, 7 at the end.



Between two marks

28–38 MIN



The end is between the 8 mark and the 9 mark. It is closer to the 9 mark.

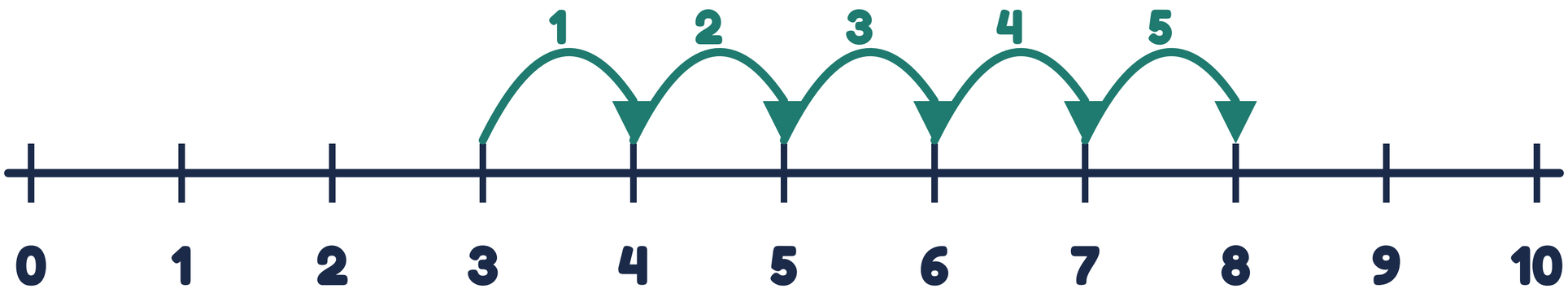
8 9

NEAREST CENTIMETRE

Choose the closer mark. This one is about 9 cm.

28–38 MIN

A line that starts at 3



The line starts at 3 and ends at 8. Count the jumps. 5 jumps. The line is 5 cm long.

8 take away 3 is 5

Your turn

28–38 MIN

Write on your paper. About 2 minutes

DRAW

Draw a line 5 cm long.

WORK OUT

A line goes from the 2 mark to the 9 mark.

How long is it?

Check: your turn

28–38 MIN

DRAW

The line starts at the 0 mark and ends at the 5 mark.

WORK OUT

9 take away 2 is 7.

7 cm

38–42 MIN

Jump the tens, hop the ones

Pretend a big jump is 10. A small hop is 1.

14 23 30 8

14 is 1 big jump and 4 small hops. Sip some water.

Workbook page 1: Start here

42–70 MIN

Questions 1 to 3. About 9 minutes. Use your ruler

S1 **In the box, draw a line 7 cm long. Start at the 0 mark.**

Your line ends at the ☐ mark.

S2 **In the box, draw a line 9 cm long. Ali's line is 4 cm long.**

Your line is _____ cm longer than Ali's line.

S3 **Measure something short from your table, such as a crayon or a rubber. Write your guess first. Then measure it in whole cm.**

(a) My guess: _____ cm

(b) I measured: _____ cm

Check page 1

42–70 MIN

S1	The line ends at the 7 mark (it starts at the 0 mark and is 7 cm long)
S2	5 cm longer (9 take away 4)
S3	Any answer that matches the thing you measured and is close to your guess. A grown-up checks.

Workbook page 2: Start here

42–70 MIN

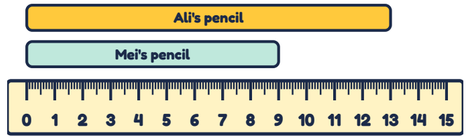
Questions 4 to 6. About 8 minutes. Use your ruler

S4 **Look at the ruler. Ali's pencil and Mei's pencil both start at the 0 mark.**

(a) Ali's pencil is _____ cm long.

(b) Mei's pencil is _____ cm long.

(c) Ali's pencil is _____ cm longer than Mei's.



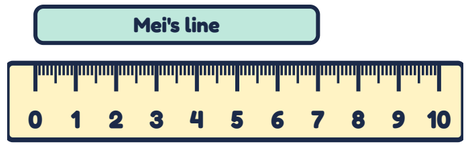
S5 **Mei's line is 5 cm long. In the box, draw a line 3 cm longer than Mei's line.**

Your line is _____ cm long.

S6 **Look at Mei's line on the ruler. Mei says it is 8 cm long.**

(a) Is Mei correct? Circle Yes or No.

(b) Her line is _____ cm long.



Check page 2

42–70 MIN

S4	13 cm; 9 cm; Ali's is 4 cm longer
S5	The second line is 8 cm long (5 + 3)
S6	No. Her line is 7 cm. She should draw on to the 8 mark

42–70 MIN

Choose your next page

Page	What it is
Think harder (page 3)	Our idea in a new setting
Go further (page 4)	Bigger puzzles
Investigation (page 5)	Does the start mark matter?
Check	Finish and check your own work

Everyone may choose. You can swap at any time.

Talk and explain

70–76 MIN

MEI AND BEN MEASURE THE SAME PENCIL

Mei says 15 cm.

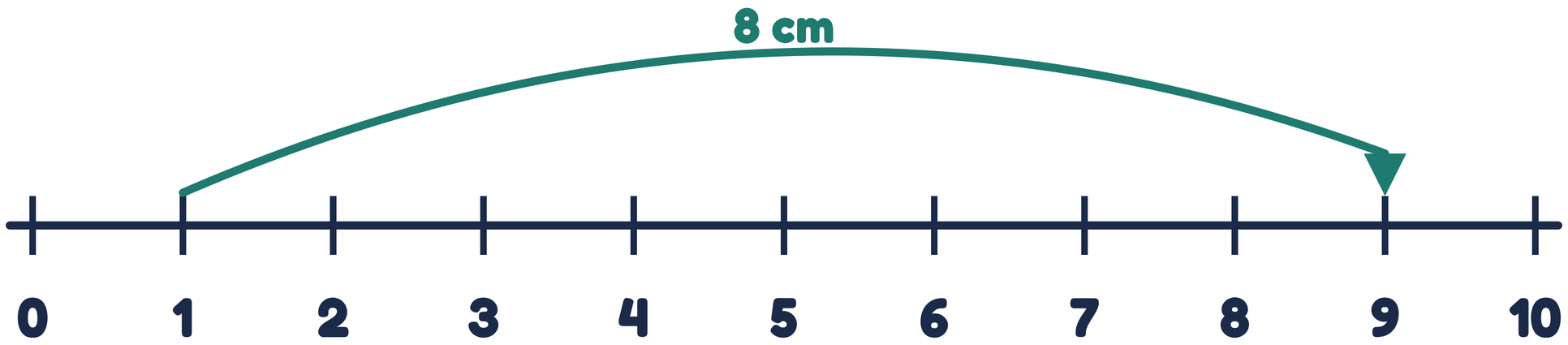
Ben says 16 cm.

Why might their answers be different? How can we check?

I think … because …

Back to the sneak-peek

70–76 MIN



No! The pencil starts at the 1 mark, so it is 8 cm long: 9 take away 1.

Exit ticket

76–82 MIN

On your own. No looking back. About 3 minutes

E1 **In the box, draw a line 6 cm long. Start at the 0 mark.**

Your line ends at the ☐ mark.

E2 **Siti's rubber ends at the 6 mark and her pencil ends at the 15 mark. Both start at the 0 mark.**

The pencil is _____ cm longer than the rubber.

E3 **A line starts at the 3 mark and ends at the 10 mark. How long is the line?**

The line is _____ cm long.

Check the exit ticket

76–82 MIN

E1	A line from the 0 mark to the 6 mark
E2	9 cm longer
E3	7 cm. 10 take away 3 is 7 (7 jumps from 3 to 10)

Look-back

82–85 MIN

Three minutes. There are no right answers. You can pass. Nobody reads it out.

Something I did well	write or draw one thing
The hardest part was …	☐ starting at the 0 mark ☐ drawing a line exactly ☐ a line that starts at another mark ☐ adding the pieces of a path ☐ something else
What helped me	☐ my ruler ☐ drawing it ☐ trying again ☐ I asked for help ☐ I was not stuck
Next time I will try	☐ drawing it ☐ using objects ☐ checking my answer ☐ asking a question
	☐ I am still stuck (my grown-up will photograph this page for my tutor) ☐ I would like to pass

Take your time. You can pass.



85–90 MIN

Well done, inquirers!

Show a grown-up one thing you found out today.

NEXT LESSON

Story Steps and Mi's Challenge Quest. Have ready: about 20 counters, your ruler, a pencil and paper, the Quest Log in its envelope (keep it closed) and a quest partner (a grown-up, or a brother or sister).



Go further: how many cuts?

OPTIONAL

A strip of paper is 10 cm long. You may fold it at the 1 mark, the 2 mark and so on, but not at the ends.

How many places can you fold? What if the strip is 20 cm long?

OPTIONAL

If we have time: pick one

Activity	What we do
Is it always true?	A line ends at the 9 mark. Is it always 9 cm long?
Make a puzzle	Draw a path with 3 straight pieces for your grown-up to measure
Teach it back	Show how to draw a line 9 cm long, in steps

Optional. Everyone may join.